

Given: Due:	Further Maths (Decision) HW 04	Name:
Chapter 1: Algorithms		
1) 42 21 15 16 35 10 31 11 27 39 (a) Use the first-fit bin packing algorithm to determine how the numbers listed above can be packed into bins of size 65 (3) (b) The list of numbers is to be sorted into descending order. Use a quick sort to obtain the sorted list. You should show the result of each pass and identify your pivots clearly. (4) (c) Use the first-fit decreasing bin packing algorithm on your ordered list to pack the numbers into bins of size 65 (3) The nine distinct numbers below are to be sorted into descending order 23 14 17 x 21 18 8 20 11 A bubble sort, starting at the left-hand end of the list, is to be used to obtain the sorted list. After the first complete pass, the list is 23 17 x 21 18 14 20 11 8 After the second complete pass, the list is 23 17 21 18 x 20 14 11 8 (d) Using this information, write down the smallest interval that must contain x. Give your answer as an inequality. (3)		

Examiners reported that a significant number of students struggled in applying the first-fit bin packing algorithm in part (a). This was mainly down to not applying the algorithm correctly. First fit is just that; students must decide if the current item under consideration will fit in the first bin rather than the most recent bin used. In this part a number of students placed the 10 in the third bin (and not the second bin) and others did not place the 27 in the third bin.

Many correct solutions were seen in part (b), but a number of students did not choose their pivots consistently, switching between middle-left and middle-right pivots during the course of the quick sort algorithm. A number of students either lost an item or changed an item during the sort, and in a small number of cases only one pivot was chosen per iteration. As stated in previous examiners' reports, students must make it clear that the sort is complete by either explicitly stating that the sort is complete or by choosing each item as a pivot or by rewriting the final list. Common errors included the items 42 and 39 being interchanged in the second pass and/or the 27 and 21 not being interchanged in the fifth pass; students should be reminded that items should remain in the order from the previous pass as they move into sub-lists. There were only a few instances where students selected the first or last items as the pivot. Pivots were usually chosen consistently although the spacing and notation on some solutions made these difficult for examiners to follow. Some students over complicated the process by insisting on using a different 'symbol' to indicate the pivots for each pass. Those students who sorted into ascending order usually remembered to reverse their list at the end to gain full credit although a number of students left their list in ascending order.

The first-fit decreasing in part (c) was well carried out with only a small minority failing to attempt this part. There were a large number of wholly correct answers. A small number performed first-fit increasing therefore scoring no marks. A small minority of students lost all three marks by placing the 27 in the 4th rather than 3rd bin (so failing to apply the algorithm at its first real test). Some students wrote totals in the bin rather than the next value. A variety of different layouts were used but in nearly all cases were easy to read and decipher.

Part (d) discriminated well with many students correctly determining that $14 < x < 17$. However, a number of students gave an answer which included the value of 17 even though the question clearly said that the numbers in the list were distinct or incorrectly assumed that x must be an integer.



Question Number	Scheme	Marks
(a)	Bin 1: <u>42</u> <u>21</u> Bin 2: <u>15</u> <u>16</u> <u>10</u> <u>11</u> Bin 3: <u>35</u> 27 Bin 4: <u>31</u> Bin 5: 39	<u>M1</u> <u>A1</u> A1 (3)
(b)	e.g. middle right 42 21 15 16 35 <u>10</u> 31 11 27 39 Pivot 10 42 21 15 16 <u>35</u> 31 11 27 39 <u>10</u> Pivot 35 42 39 <u>35</u> 21 15 16 <u>31</u> 11 27 <u>10</u> Pivot 39, 31 42 <u>39</u> <u>35</u> <u>31</u> 21 15 <u>16</u> 11 27 <u>10</u> Pivots (42) 16 42 <u>39</u> <u>35</u> <u>31</u> 21 <u>27</u> <u>16</u> 15 <u>11</u> <u>10</u> Pivots 27, 11 42 <u>39</u> <u>35</u> <u>31</u> <u>27</u> 21 <u>16</u> 15 <u>11</u> <u>10</u> Sort complete	M1 A1 A1ft A1 (4)
(c)	Bin 1: <u>42</u> <u>21</u> Bin 2: <u>39</u> <u>16</u> 10 Bin 3: <u>35</u> <u>27</u> Bin 4: <u>31</u> <u>15</u> 11	<u>M1</u> <u>A1</u> A1 (3)
(d)	$14 < x < 17$	B3,2,1,0 (3)
		13 marks

Notes for Question

a1M1: First five items placed correctly and at least eight values placed in bins - condone cumulative totals for M1 only (the boxed values)

a1A1: First eight items placed correctly (the boxed and underlined values)

a2A1: CSO (so no additional/repeated values)

b1M1: Quick sort, pivot, p, chosen (must be choosing middle left or right – choosing first/last item as the pivot is M0). After the first pass the list must read (values greater than the pivot), pivot, (values less than the pivot). **If only choosing one pivot per iteration then M1 only** – Bubble sort is not a MR and scores M1 only for 42 21 16 35 15 31 11 27 39 10 (for left to right) or 42 39 21 15 16 35 10 31 11 27 (for right to left)

b1A1: First two passes correct and next pivots chosen correctly for the third pass (but the third pass does not need to be correct) – so they must be choosing (if middle right) pivot values of 39 and 31 for the third pass or (if middle left) a pivot value of 16

b2A1ft: Third and fourth passes correct (follow through from their second pass and choice of pivots). They do not need to be choosing a pivot for the fifth pass for this mark

b3A1: CSO (correct solution only – all previous marks in this part must have been awarded) including a fifth pass in which the 27 and 11 (if middle right) or 21 (if middle left) is used as a pivot and a 'sort complete' - this could be shown by the final list being re-written or 'sorted' statement or each item being used as a pivot (which would therefore mean that the final list would have been written twice)

c1M1: Must be using 'sorted' list in descending order. First five items placed correctly and at least eight values placed in bins – condone cumulative totals for M1 only (the boxed values)

c1A1: First eight items placed correctly (the underlined and boxed values)

c2A1: CSO (so no additional/repeated values)

SC for part (c) – if 'sorted' list is incorrect from part (b) and M0 would be awarded in (c) then award M1 only in (c) for their first eight items correctly placed – by 'incorrect' they can have only one 'error' – an 'error' is one missing number, one extra number, or one number incorrectly placed. Allow full marks in (c) if a correct list is used in (c) even if the list is incorrect at the end of (b)



Question Number	EXAM PAPERS PRACTICE Scheme	Marks
IN (d) mark their final attempt only – if in doubt please send to review d1B1: Any inequality in x linked with either 14 or 17 (allow in words or using set notation) or an indication of an interval with 14 and 17 but with no inequality sign (or even x) present d2B1: Any inequality in x linked with 14 and 17 (allow in words) or using set notation (14, 17) d3B1: CAO (allow as two separate inequalities: $x > 14, x < 17$ (allow without a comma separating the two inequalities), accept $x > 14$ and $x < 17$ but not $x > 14$ or $x < 17$) - if defined using words only then do not award this mark SC: award B1B1B0 in (d) for $15 \leq x \leq 16$ or $[15,16]$ (no marks for any strict inequalities with 15/16) Misreads - If the candidate has misread a number at the start of (a), so genuinely miscopy a number then mark the whole of (a), (b) and (c) as a misread – removing the last two A marks earned. This gives a maximum of 8 marks in total for these three parts - If they have used the correct numbers at any point in (a) and then use an incorrect number in (b) (say 24 instead of 42) from the beginning of the sort or misread one of their own numbers during (b) then count it as one 'error' in (b) (so they will lose at least the final A mark in (b) but should be able to gain at least the M mark and the follow through A mark) – then mark (c) according to the SC above. More than one 'error' in (b) loses all subsequent A marks in (b)		

Sorting list into ascending order in (b)

- If the candidate sorts the list into ascending order and reverses the list in this part then this can score full marks in (b)
- If the list is not reversed in (b) then mark as a misread (so remove the last two A marks earned in (b)). If the list is reversed at the start of (c) but not in (b) then still treat this as a misread. If the list is in ascending order in (b) award no marks for first-fit increasing in (c). If the candidate says that the list needs reversing in (b) but does not actually show the reversed list in (b) then remove the final A mark

Middle left for (b):

42	21	15	16	35	10	31	11	27	39
42	39	35	21	15	16	10	31	11	27
42	39	35	21	15	16	31	11	27	10
42	39	35	21	31	27	16	15	11	10
42	39	35	31	21	27	16	15	11	10
42	39	35	31	27	21	16	15	11	10

Pivot 35
Pivots 42, 10
Pivots (39) 16
Pivots 31, 15
Pivot 21 (11)
Sort complete

Ascending middle right

42	21	15	16	35	10	31	11	27	39
10	42	21	15	16	35	31	11	27	39
10	21	15	16	31	11	27	35	42	39
10	21	15	16	11	27	31	35	39	42
10	15	11	16	21	27	31	35	39	42
10	11	15	16	21	27	31	35	39	42

Ascending middle left

42	21	15	16	35	10	31	11	27	39
21	15	16	10	31	11	27	35	42	39
10	21	15	16	31	11	27	35	39	42
10	15	11	16	21	31	27	35	39	42
10	11	15	16	21	27	31	35	39	42
10	11	15	16	21	27	31	35	39	42

(Sort complete)

2)

59 45 18 55 47 11 63 17 15 42

(a) The list of numbers above is to be sorted into descending order. Perform a quick sort to obtain the sorted list. You should show the result of each pass and identify your pivots clearly.

(4)

The numbers in the list represent the lengths, in cm, of some pieces of copper wire. The copper wire is sold in one metre lengths.

(b) Use the first-fit decreasing bin packing algorithm to determine how these pieces could be cut from one

metre lengths. (You should ignore wastage due to cutting.)

(3)

(c) Determine whether your solution to (b) is optimal. Give a reason for your answer.

(2)

In part (a), most students selected middle-right pivots and many were able to carry out the sort correctly. Errors cropped up in the ordering of the sublists after the second (and subsequent) passes. The most common occurrence of this tended to be that the '55' and '63' were interchanged after the second pass. Other errors included failing to select the '42' as a pivot for the fifth pass. As stated in previous examiners' reports even though after the fourth pass the list 'appears' to be in order, as demanded by the algorithm, a final pass through the list is required. Very occasionally, students selected only one pivot for each iteration or failed to sort the list into (values greater than the pivot), (the pivot), (values less than the pivot) after the first iteration. There were only a few instances where students selected the first or last items as the pivot. Pivots were usually chosen consistently although the spacing and notation on some solutions made these difficult for examiners to follow. Some students over-complicated the process by insisting on using a different 'symbol' to indicate the pivots for each pass. Those students who sorted into ascending order usually remembered to reverse their list at the end to gain full credit, although a number of students left their list in ascending order.

The first-fit decreasing in part (b) was well carried out with only a small minority failing to attempt this part. There were a large number of fully correct answers. A small number performed first-fit increasing therefore scoring no marks. A small minority of students lost all the marks by placing the '45' in the 4th rather than 3rd bin (so failing to apply the algorithm at its first real test). Some students wrote totals in the bin rather than the next value. A variety of different layouts were used but in nearly all cases were easy to read and decipher.

Part (c) was answered very well by the majority of students. Most errors that were made were avoidable – arithmetic errors or failing to answer the question in stating that their solution was optimal. A minority of students attempted an alternative to finding the lower bound, most of these referred to the total wastage relative to the bin size, and these approaches were rarely successful. Students need to make sure their answers are clear – some lost a mark in part (c) as, although they correctly found that the lower bound was 4, they did not explicitly refer to their solution in part (b) as therefore being optimal.

Question Number	Scheme	Marks
(a)	e.g. using middle right 59 45 18 55 47 11 63 17 15 42 59 45 18 55 47 63 17 15 42 11 59 55 63 47 45 18 17 15 42 11 59 63 55 47 45 18 42 17 15 11 63 59 55 47 45 42 18 17 15 11 63 59 55 47 45 42 18 17 15 11 pivot 11 pivot 47 pivot 55 17 pivot 63 18 (15) pivot (59) 42 (sort complete)	M1 (quick) A1 (2 passes + choice of pivot for the 3 rd) A1ft (3 rd and 4 th passes correct) A1 (CSO) (4)
(b)	Bin 1: 63 18 17 Bin 2: 59 15 11 Bin 3: 55 45 Bin 4: 47 42	M1 A1 A1 (3)
(c)	$\frac{372}{100} = 3.72$ so yes the solution in (b) is optimal	M1 A1 9 marks (2)

Notes for Question

a1M1: Quick sort, pivot, p, chosen (must be choosing middle left or right – choosing first/last item as pivot is M0). After the first pass the list must read (values greater than the pivot), pivot, (values less than the pivot). **If only choosing one pivot per iteration then M1 only** – Bubble sort is not a MR and scores M1 only for 59 45 55 47 18 63 17 15 42 11 (for left to right) or 63 59 45 18 55 47 11 42 17 15 (for right to left)

a1A1: First two passes correct **and** next pivots chosen correctly for third pass (but third pass does not need to be correct) – so they must be choosing (if middle right) pivot values of 55 and 17 for the third pass or (if middle left) pivot values of 59 and 17

a2A1ft: Third and fourth passes correct (follow through from their second pass and choice of pivots). They do not need to be choosing a pivot for the fifth pass for this mark

a3A1: CSO (correct solution only – all previous marks in this part **must** have been awarded) including a fifth pass in which the 42 (if middle right) or 45 (if middle left) is used as a pivot (not just stated as a pivot)

b1M1: **Must be using 'sorted' list** in descending order. First five items placed correctly and at least eight values placed in bins – condone cumulative totals for M1 only (the underlined values)

b1A1: First eight items placed correctly (the underlined **and** boxed values)

b2A1: CSO

SC for part (b) – if 'sorted' list is incorrect from part (a) and M0 would be awarded in (b) then award M1 only in (b) for **their** first eight items correctly placed – by 'incorrect' they can have only **one** error, e.g. one missing number, one extra number, or one number incorrectly placed

c1M1: Attempt to find lower bound $(372 \pm 63) / 100$ (a value of 3.72 seen with no working can imply this mark) or any argument based on the four largest values

c1A1: CSO – correct calculation seen or 3.72 **and** a conclusion – accept 'yes' as a minimum conclusion – however, '4 is the optimal number of bins' (or equivalent) with no reference to the solution in (b) is A0. For those using the four largest values argument they must clearly explain why two of these values cannot be placed in a bin e.g. the sum of any two of 63, 59, 55, 47 is greater than 100 so no two can be placed in a bin



Question Number	Scheme	Marks
For part (a) using middle left as pivot		
59 45 18 55 <u>47</u> 11 63 17 15 42	pivot 47	
59 <u>55</u> 63 <u>47</u> 45 18 <u>11</u> 17 15 42	pivot 55 11	
<u>59</u> 63 <u>55</u> <u>47</u> 45 18 <u>17</u> 15 42 <u>11</u>	pivot 59 17	
63 <u>59</u> <u>55</u> <u>47</u> 45 <u>18</u> 42 <u>17</u> 15 <u>11</u>	pivot (63) 18 (15)	
63 <u>59</u> <u>55</u> <u>47</u> <u>45</u> 42 <u>18</u> <u>17</u> 15 <u>11</u>	pivot 45	
63 <u>59</u> <u>55</u> <u>47</u> <u>45</u> 42 <u>18</u> <u>17</u> 15 <u>11</u>	(sort complete)	
Misreads		
- If the candidate has misread a number at the start of (a), so genuinely miscopy a number then mark the whole question as a misread – removing the last two A marks earned. This gives a maximum of total of 7 marks - If the candidate starts the sort with the correct numbers in (a) but they misread their own numbers (so they have a copying error) during the sort then count this as an error in (a) but mark (b) and (c) as a misread. If they restart in (b) and (c) with the correct list of numbers then this is fine for full marks		
Sorting list into ascending order in (a)		
- If the candidate sorts the list into ascending order and reverses the list in this part then this can score full marks in (a) - If the list is not reversed in (a) then mark as a misread (so remove the last two A marks earned in (a)). If the list is reversed at the start of (b) but not in (a) then still treat this as a misread. If the list is in ascending order in (b) award no marks for first-fit increasing. If the candidate says that the list needs reversing in (a) but does not actually show the reversed list in (a) then remove the final A mark		
Ascending (middle right)		
59 45 18 55 47 <u>11</u> 63 17 15 42	(11)	M1
<u>11</u> 59 45 18 55 <u>47</u> 63 17 15 42	(47)	
<u>11</u> 45 18 <u>17</u> 15 42 <u>47</u> 59 <u>55</u> 63	(17, 55)	A1
<u>11</u> 15 <u>17</u> 45 <u>18</u> 42 <u>47</u> <u>55</u> 59 <u>63</u>	((15), 18, 63)	
<u>11</u> 15 <u>17</u> <u>18</u> 45 <u>42</u> <u>47</u> <u>55</u> 59 <u>63</u>	(42, (59))	A1ft
<u>11</u> 15 <u>17</u> <u>18</u> <u>42</u> 45 <u>47</u> <u>55</u> 59 <u>63</u>		A1 CSO + 'sort complete' statement
Ascending (middle left)		
59 45 18 55 <u>47</u> 11 63 17 15 42	(47)	M1
45 18 <u>11</u> 17 15 42 <u>47</u> 59 <u>55</u> 63	(11, 55)	
<u>11</u> 45 18 <u>17</u> 15 42 <u>47</u> <u>55</u> <u>59</u> 63	(17, 59)	A1
<u>11</u> 15 <u>17</u> 45 <u>18</u> 42 <u>47</u> <u>55</u> <u>59</u> 63	((15), 18, (63))	
<u>11</u> 15 <u>17</u> <u>18</u> <u>45</u> 42 <u>47</u> <u>55</u> <u>59</u> 63	(45)	A1ft
<u>11</u> 15 <u>17</u> <u>18</u> 42 <u>45</u> <u>47</u> <u>55</u> <u>59</u> 63		A1 CSO + 'sort complete' statement

3)

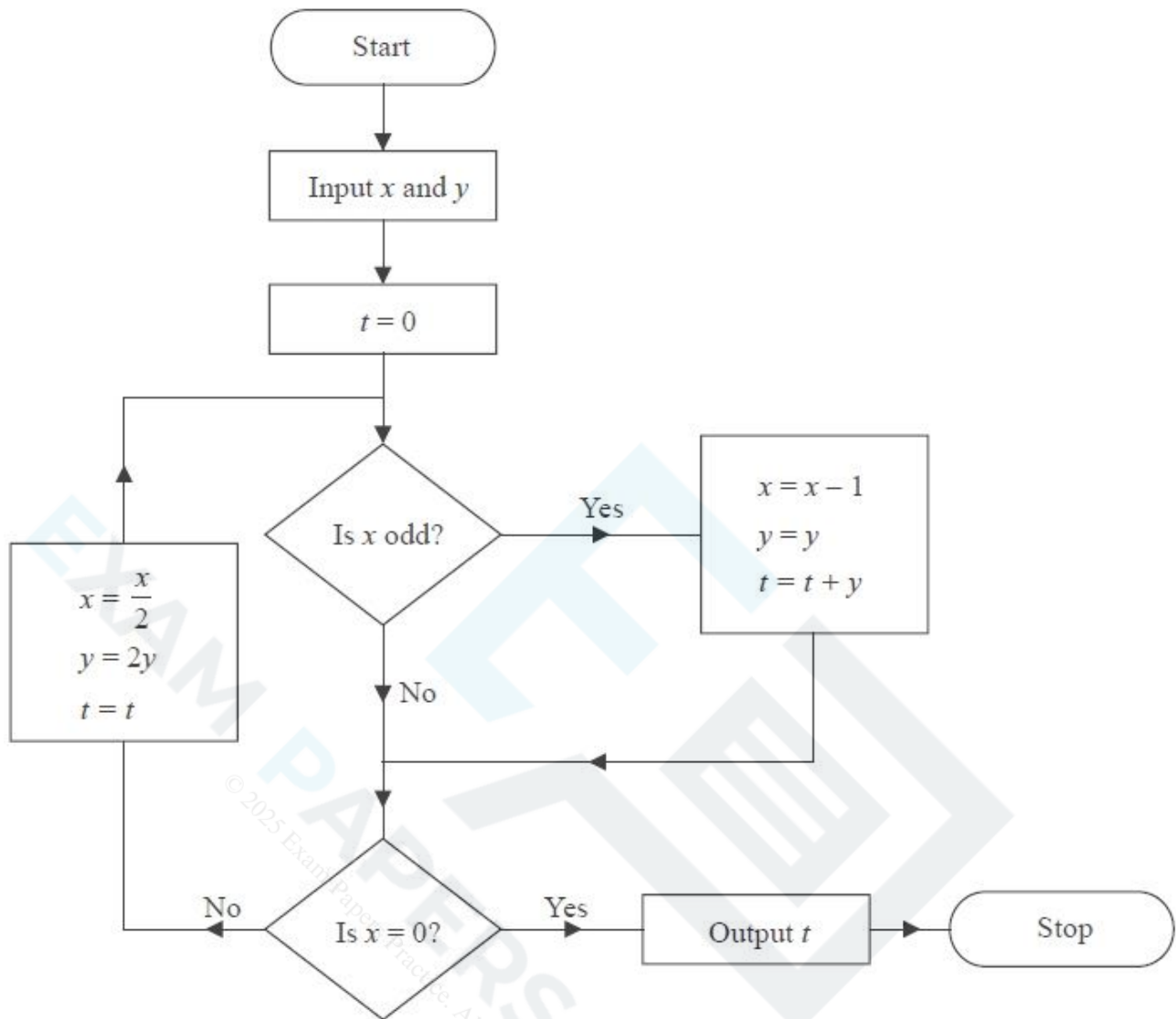


Figure 4

An algorithm is described by the flow chart shown in Figure 4.

Given that $x = 27$ and $y = 5$,

(a) complete the table in the answer book to show the results obtained at each step when the algorithm is applied. Give the final output.

(4)

The numbers 122 and $\frac{1}{2}$ are to be used as inputs for the algorithm described by the flow chart.

(b) (i) State, giving a reason, which number should be input as x .

(ii) State the output.

(3)

The majority of students showed a sound understanding of applying the given flow chart, scoring 3 or full marks in part (a). However a few lost all marks in part (a) for failing to fill in the first row and instead starting with the second row. Arithmetic errors were surprisingly rare. Extra "yes" and "no" entries were quite common in the fourth and fifth columns and were penalised with the final mark in this part.

Many scored the first B1 mark in part (b) for identifying that x should be 122, but then lost the next mark struggling to correctly explain why. The simplest explanation that x must be an integer, was spotted by some,

as was "2 is neither odd nor even". However many simply focused on the 2 stating that the algorithm would not terminate, or x would never = 0, but failing to explain why.

The correct answer of 61 was often seen for the final mark in part (b). Many realised that it was simply the product of the two inputs, but some wasted time working through the algorithm for a second time with the two values of 122 and 2.

Question Number	Scheme	Marks																																																																		
(a)	<table><tr><th>x</th><th>y</th><th>t</th><th>Is x odd?</th><th>Is $x = 0$?</th><th></th></tr><tr><td>27</td><td>5</td><td>0</td><td>Yes</td><td></td><td></td></tr><tr><td>26</td><td>(5)</td><td>5</td><td></td><td>No</td><td>Row 1</td></tr><tr><td>13</td><td>10</td><td>(5)</td><td>Yes</td><td></td><td>Row 2</td></tr><tr><td>12</td><td>(10)</td><td>15</td><td></td><td>No</td><td>Row 3</td></tr><tr><td>6</td><td>20</td><td>(15)</td><td>No</td><td>No</td><td>Row 4</td></tr><tr><td>3</td><td>40</td><td>(15)</td><td>Yes</td><td></td><td>Row 5</td></tr><tr><td>2</td><td>(40)</td><td>55</td><td></td><td>No</td><td>Row 6</td></tr><tr><td>1</td><td>80</td><td>(55)</td><td>Yes</td><td></td><td></td></tr><tr><td>0</td><td>(80)</td><td>135</td><td></td><td>Yes</td><td></td></tr><tr><td></td><td></td><td>(135)</td><td></td><td></td><td></td></tr></table> Output = 135	x	y	t	Is x odd?	Is $x = 0$?		27	5	0	Yes			26	(5)	5		No	Row 1	13	10	(5)	Yes		Row 2	12	(10)	15		No	Row 3	6	20	(15)	No	No	Row 4	3	40	(15)	Yes		Row 5	2	(40)	55		No	Row 6	1	80	(55)	Yes			0	(80)	135		Yes				(135)				M1 (3 rows + 1 st correct) A1 (2 nd and 3 rd rows correct) A1 (4 th , 5 th and 6 th rows correct) A1 (CSO) (4)
x	y	t	Is x odd?	Is $x = 0$?																																																																
27	5	0	Yes																																																																	
26	(5)	5		No	Row 1																																																															
13	10	(5)	Yes		Row 2																																																															
12	(10)	15		No	Row 3																																																															
6	20	(15)	No	No	Row 4																																																															
3	40	(15)	Yes		Row 5																																																															
2	(40)	55		No	Row 6																																																															
1	80	(55)	Yes																																																																	
0	(80)	135		Yes																																																																
		(135)																																																																		
(b)(i)	x must be a (positive) integer and therefore $x = 122$	B1 DB1																																																																		
(ii)	61	B1 (3) 7 marks																																																																		

Notes for Question

Candidates may write each changed value/statement in a new row which is fine. Assume that each row begins and ends when a value in x is changed. For example, the values in row 1 in the table above consists of the x values going from the 26 to the 13

a1M1: At least three rows of cells in columns x , y and t completed with a correct first row (so 26 for x and 5 for t)

a1A1: CAO – second and third rows correct (for just the columns in x , y and t)

a2A1: CAO – fourth, fifth and sixth rows correct (for just the columns in x , y and t)

a3A1: CSO – including the output of 135 either on the given line in the answer book or clearly stated in the table but it must be absolutely clear that the output is the final t value (no bod). Furthermore, all 'yes' and 'no' comments must be present in the 4th and 5th columns with no additional/incorrect 'yes' or 'no'

bi1B1: x must be 122 and any attempt at a reason

bi2DB1: Dependent on previous B mark (so B0B1 is not possible) – 122 and a correct valid reason – e.g. x must be an integer/whole number or $\frac{1}{2}$ is not odd or even or if you input $\frac{1}{2}$ then you can never get to $x = 0$ when halving, etc. Just saying that the algorithm 'won't work' or that the algorithm 'will get stuck in a loop' or 'not terminate' is not sufficient for this second mark neither is the argument of subtracting 1 from a $\frac{1}{2}$. It must be clear why the algorithm won't output a value for t with $x = \frac{1}{2}$ - so essentially there needs to be some indication of why x will never become 0. Furthermore, just saying that x will never reach 0 is insufficient – we need an indication of why $x = 0$ is not possible with a starting value of $x = \frac{1}{2}$

bii3B1: CAO

4)

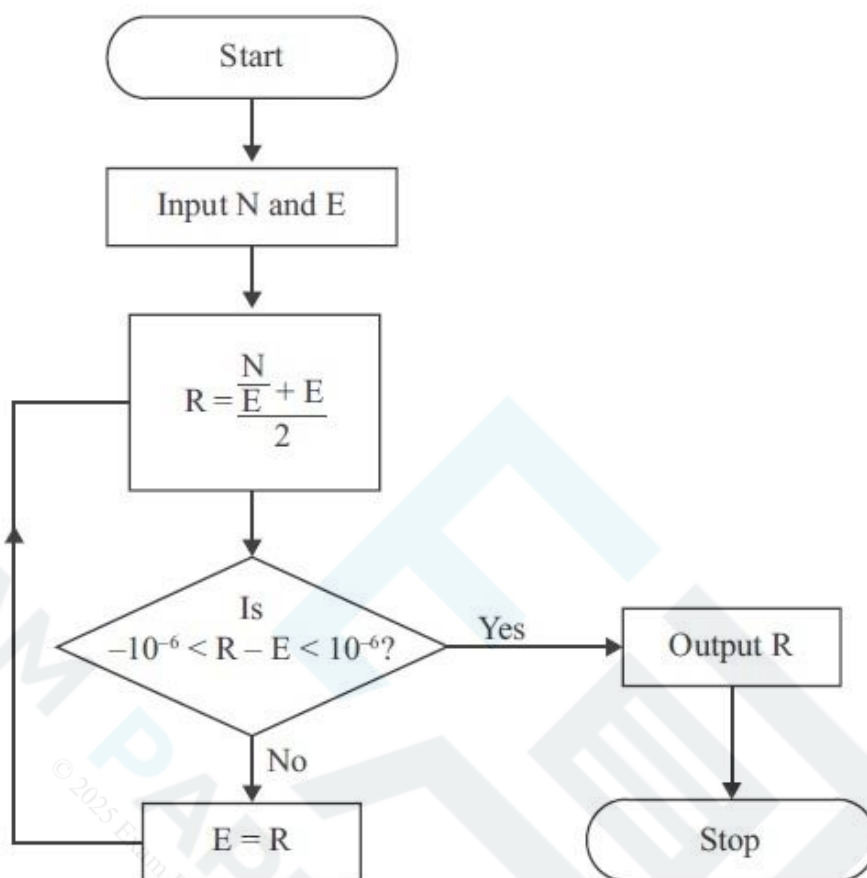


Figure 1

Hero's algorithm for finding a square root is described by the flow chart shown in Figure 1.

Given that $N = 72$ and $E = 8$,

(a) use the flow chart to complete the table in the answer book, working to at least seven decimal places when necessary. Give the final output correct to seven decimal places.

(4)

The flow chart is used with $N = 72$ and $E = -8$,

(b) describe how this would affect the output.

(1)

(c) State the value of E which cannot be used when using this flow chart.

(1)

This first question proved to be a good discriminator and gave rise to a good spread of marks. The mode was full marks gained by 23.8% of the candidates, 22.2% of the candidates scored 2 or fewer marks. (a) was completed with varying degrees of success. Most candidates could make a start on following the flow chart and usually had the first two rows of the table completed correctly. The main issue was the inability of candidates to tell whether the difference in the values of R and E fell between the two boundaries of 10^{-6} and -10^{-6} and therefore many stopped the algorithm in the wrong place (usually on the third row) or continued for many more rows than required.

Common errors in this part were not giving answers to a sufficient number of decimal places (7 were asked for in the question) or truncating their answers rather than rounding (it was popular to see 8.485281374 become 8.4852813). The majority of candidates were aware that they needed to state their final output once



they had achieved a 'yes' in the last column although many did not use the pre-printed line below the table; many opted for putting their output value under their final R value or on the next page of the answer booklet. A number of candidates instead of outputting the final value for R instead gave the value of $R - E$ as the final output. Many candidates had the 'correct' final output of $\sqrt{72}$ even though this did not follow from their working. For the final mark in this part, examiners could only follow through from the candidates working as the nature of the algorithm (for finding the square root of a number) was described in the question.

In (b) all that was required in this part was to state that the output would now be negative (or the more complete answer that the algorithm would now find the negative square root). Some candidates were convinced that there would be no affect on the output and some described what would happen at each stage of the algorithm rather than at the final output stage.

(c) was very well answered with the majority of candidates giving the correct answer that E could not be zero. A number of candidates went on to explain why E could not be zero and it was also common to see answers which implied that E could not be negative or take the value of 72.

Question Number	Scheme	Marks																				
(a)	<table><tr><th>N</th><th>E</th><th>R</th><th>Qn</th></tr><tr><td>72</td><td>8</td><td>8.5</td><td>N</td></tr><tr><td></td><td>8.5</td><td>8.485 294 118</td><td>N</td></tr><tr><td></td><td>8.485 294 118</td><td>8.485 281 374</td><td>N</td></tr><tr><td></td><td>8.485 281 374</td><td>8.485 281 374</td><td>Y</td></tr></table> Output is R = 8.485 281 4	N	E	R	Qn	72	8	8.5	N		8.5	8.485 294 118	N		8.485 294 118	8.485 281 374	N		8.485 281 374	8.485 281 374	Y	M1 A1 A1 A1ft (4)
N	E	R	Qn																			
72	8	8.5	N																			
	8.5	8.485 294 118	N																			
	8.485 294 118	8.485 281 374	N																			
	8.485 281 374	8.485 281 374	Y																			
(b)	We would get a negative output for R/ We would get the negative square root	B1 (1)																				
(c)	E cannot be zero	B1 (1)																				
	Notes a1M1: At least two rows of cells in just E and R completed. a1A1: CAO first two rows correct giving exact values or awrt 7dp (the exact second value for R is $\frac{577}{68}$). a2A1: CAO third and fourth rows awrt 7dp a3A1ft: Output for R must follow through from their final value for R awrt 7dp – candidate must have answered ‘yes’ to score this mark. Output either on the answer line (or on the second page) or stated in the table but must be in the column for R below the row which contains ‘yes’. Condone N = 72 on each row and entries appearing on separate rows throughout for full marks. Allow e.g. ticks/crosses etc. for yes/no. b1B1: Mention of ‘negative’ scores B1 however do not accept incorrect statements but bod that ‘negative’ only is implicitly describing the effect on the output. Accept ‘other square root’. c1B1: CAO (nothing/null etc. scores B0). Condone E = 0.	Total 6 marks																				